

TelecomGPT & XYZ TNIC

RCA Dashboard

Full Implementation Overview
Manager Handout

Repository: github.com/aqil2020in/telecomgpt

Products: TelecomGPT (chat AI) + XYZ TNIC (Network Intelligence Copilot)

Demo cells: XYZ401-XYZ410 | Synthetic OSS-shaped datasets

1. Executive Summary

TelecomGPT is a domain-specific multi-agent AI platform for 5G/LTE RF and network operations. The monorepo contains two related products plus shared analytics tooling. The XYZ TNIC (Telecom Network Intelligence Copilot) RCA Dashboard is a Streamlit multipage application that visualizes network health and runs explainable, rule-based root cause analysis across 17 domain pages.

Products in the monorepo

Product	Location	Purpose
TelecomGPT Chat	backend/ + frontend/	LangGraph chat + embedded RCA
XYZ TNIC	xyz_tnic/	Standalone RCA API + Streamlit dashboard
Analytics	analytics/	Generic CSV/log charts (optional)

Production deployment

- Chat UI: Vercel (telecomgpt.vercel.app)
- API: Render (telecomgpt.onrender.com)
- TNIC Dashboard: Streamlit local/Docker demo (port 8502)

2. Architecture

The same RCA engine exists in two places: xyz_tnic/tnic/ (standalone, full product) and backend/tnic/ (embedded in TelecomGPT chat via bridge.py). Both share agents, rules, datasets, and the Master RCA Orchestrator.

Data flow (preloaded demo path)

- CSV files on disk: /workspace/datasets/*.csv (15 synthetic files)
- registry.py resolves dataset directory (TNIC_DATASETS_DIR env override)
- loaders.py reads CSVs with @lru_cache (one read per Streamlit process)
- kpi_service.py merges all sources into compute_cell_kpis(cell_id)
- dashboard_utils.py exposes cell_kpis(), domain DataFrames, run_agent()
- dashboard/pages/*.py renders metrics, charts, and agent findings

3. How the RCA Dashboard Was Built

Technology

Built with Streamlit multipage convention. Entry point: xyz_tnic/dashboard/app.py. Each file in dashboard/pages/ becomes a sidebar page automatically. Numeric prefixes (2_, 3_, ...) control sort order. No custom router required.

Home page (app.py) sections

Section	Module	Purpose
Executive KPIs	app.py	Cluster health, worst cells, charts
Data Sources	data_sources_section.py	Page to CSV mapping table
AI RCA Workflow	rca_workflow_section.py	Multi-agent workflow explainer
NPI Copilot	npi_copilot_section.py	NPI validation + agent monitor
Dataset Simulation	dataset_simulation_section.py	Upload/simulate RCA

Sidebar navigation

Global Focus cell selectbox (XYZ401-XYZ410) in sidebar. Streamlit auto-discovers 17 domain pages plus home. Shared logic lives in dashboard_utils.py.

4. Standard Page Pattern

Every domain page (Handover, RLF, VoNR, etc.) follows the same 3-layer pattern:

- Layer 1 - Metrics: cell_kpis(cell_id) from merged KPI service
- Layer 2 - Visualization: domain DataFrame filtered by cell (charts + tables)
- Layer 3 - RCA: run_agent(domain, cell_id) returns rule engine findings

Exceptions: RCA Report runs full MasterRCAOrchestrator; Assurance Hub runs all 12 agents; Upload uses ingest pipeline; RF Coverage uses geospatial Plotly maps.

5. Sidebar Pages Inventory

Page	Data Source	Agent
Handover	handover_events_enriched.csv	HOAgent
RLF	rlf_events.csv	RLFAgent
Call Drops	call_drop_events.csv	CallDropAgent
RACH	rach_events.csv + PM	RACHAgent
Throughput	throughput_metrics.csv	ThroughputAgent
Beamforming	PM + synthetic beams	BeamformingAgent
VoNR	vonr_sessions.csv	VoNRAgent
ANR	anr_events.csv	ANRAgent
Config Audit	cell_configuration.csv	ConfigAuditAgent
gNB Syslog	gnb_syslog.csv	GNBSyslogAgent
Alarm Correlation	alarm_events.csv	AlarmAgent
UE Protocol	ue_protocol_trace.csv	UEProtocolAgent
RF Coverage	geospatial RF CSV	RFCoverageAgent
RCA Report	All merged KPIs	MasterRCAOrchestrator
Assurance Hub	All assurance CSVs	All 12 agents
Upload	User files	Dynamic RCA pipeline

6. RCA Agent Pipeline

Specialist agents (12+)

File: xyz_tnic/tnic/agents/specialists.py. Each agent wraps a deterministic rule engine (tnic/rules/*.py). Returns rule_id, probable_cause, confidence, evidence, and recommended_actions. No LLM required for core RCA.

Master RCA Orchestrator

File: xyz_tnic/tnic/orchestrator/rca_orchestrator.py. Steps:

- 1. detect_issue_type() - keyword classifier
 - 2. ORCHESTRATION_MAP selects 8-12 specialist agents
 - 3. Fan-out: each agent runs analyze(kpis, query)
 - 4. Primary rule engine pass + call-drop classifier
 - 5. master_rca.py enrichment (coverage correlation, workflows)
 - 6. rank_findings() with domain boost (+10%) and classifier boost (+15%)
 - 7. Output: ranked causes, actions, health score, knowledge graph
- 28 NOC-grade workflows defined in rca_catalog.py (coverage hole, ping-pong, VoNR drop, etc.)

7. TelecomGPT Integration

File: backend/tnic/bridge.py. When users ask RCA-style questions in the Next.js chat, `looks_like_tnic_rca_query()` detects intent, `run_tnic_rca()` loads KPIs from datasets, runs MasterRCAOrchestrator, and returns markdown plus agent trace. Same engine, different UI (chat vs Streamlit).

8. Upload & Dynamic RCA Path

- User uploads file via Upload page or home simulation
- `file_classifier.py` identifies format (CSV, log, Excel, zip)
- `normalization_engine.py` maps to canonical event schema
- Stored as `events.jsonl` under `xyz_tnic/data/uploads/<id>/`
- `dynamic_rca.py` merges upload KPIs with bundled cell KPIs
- MasterRCAOrchestrator produces full RCA report

Upload is additive - does not replace preloaded CSVs for sidebar pages.

9. Key Design Decisions

- CSV-first demo with OSS-shaped schema - swap data via loaders, not orchestration
- Rule-based RCA core, OpenAI optional for narrative reports only
- Single KPI merge layer (`kpi_service.py`) - no hard-coded UI numbers
- Consistent page template - metrics + charts + agent findings
- Two products, one engine - standalone TNIC + TelecomGPT embed
- Assurance datasets first-class - VoNR, ANR, syslog, alarms, UE trace

10. How to Run & Demo

From repo root:

```
cd xyz_tnic
export TNIC_DATASETS_DIR=/workspace/datasets
streamlit run dashboard/app.py --server.port 8502
```

Recommended demo flow for management:

1. Home - cluster health, data sources, RCA workflow explainer
2. Handover - pick XYZ401, show KPIs + HO Agent findings
3. VoNR or RLF - domain-specific RCA
4. RCA Report - full multi-agent orchestration
5. Assurance Hub - all 12 agents on one cell
6. Upload/Simulation - live ingest to Master RCA

11. Documentation References

Document	Path
Full architecture	docs/ARCHITECTURE.md
Dashboard data flow	docs/TNIC_DASHBOARD_DATA_FLOW.md
Handover RCA deep dive	docs/RCA_AGENT_END_TO_END_HANDOVER.md
Manager demo script	docs/DEMO_MANAGER.md
Platform overview PDF	docs/XYZ Telecom TNIC.pdf
Cloud agent commands	AGENTS.md